

Xiaoyang Bai, Ph.D.

Department of Electrical and Computer Engineering, HKU
xybai@hku.hk | (+852) 93522980 | andrewbxy.github.io

RESEARCH INTERESTS

Physically-grounded 3D perception and reconstruction with **unconventional visual modalities** (event stream, polarization image, etc.), for real-world applications under **challenging scenarios**.

EDUCATION

College of Engineering, University of Illinois at Urbana-Champaign (UIUC) 2018 - 2024

- **Ph.D. in Computer Science**
- **Advisory Committee:** Alexander Schwing, David Forsyth (*ACM/IEEE Fellow*), Viktor Gruev
- **Thesis:** *Underwater Polarization-based Underwater Geolocalization with Neural Network Models*

College of Letter and Science, University of California, Berkeley 2014 - 2018

- B.A. in Computer Science, Applied Math, Cognitive Science and Linguistics

ACADEMIC APPOINTMENTS

Postdoctoral Fellow, WeLight Lab, The University of Hong Kong 2024 - present

- 3D acquisition & reconstruction with unconventional visual modalities
- **Supervisors:** Prof. Yifan (Evan) Peng, Prof. Edmund Y. Lam

Graduate Research Assistant, University of Illinois at Urbana-Champaign 2018 - 2024

- Polarization-based underwater localization

Research Scientist Intern, Amazon Studio LLC 2022

- Autoregressive video prediction

AI Research Intern, Apple Inc. 2020

- Sequential data generation and prediction

RESEARCH GRANTS

Co-I, Innovation and Technology Support Programme (ITSP), ITC (HKSAR) 2025 - 2027

- Title: 3D Reconstruction of Complex Dynamic Scenes with Event-RGB Multi-modal Fusion

PI, State Key Lab of CAD & CG Open Topic Research Scheme, ZJU 2024 - 2025

- Title: Real-time 3D Scene Reconstruction Based on Egocentric Event Streams

PROFESSIONAL SERVICES

Conference Organizer

- **Program Manager of XR**, ACM SIGGRAPH Asia 2025
- **Program Committee of Emerging Technology**, ACM SIGGRAPH Asia 2026
- **Technical Paper Committee**, IEEE International Conference on Computational Imaging (ICCP) 2026

- **Technical Paper Committee**, International Conference on Extended Reality (ICXR) 2025

Journal & Conference Reviewer

- ACM Transactions on Graphics (TOG)
- ACM SIGGRAPH & SIGGRAPH Asia
- IEEE Transactions on Visualization and Computer Graphics (TVCG)
- IEEE International Symposium on Mixed and Augmented Reality (ISMAR)
- IEEE Conference on Virtual Reality (VR)
- IEEE International Conference on Computational Imaging (ICCP)
- Computer Visual Media Journal (CVMJ)
- Computer & Graphics
- Computer Graphics Forum

PUBLICATIONS

(Award Candidate, 0.47% of total submissions) PhyGaP: Physically-grounded Gaussians with polarization cues. Jiale Wu, **Xiaoyang Bai**^{*}, Zongqi He, Weiwei Xu, Yifan Peng^{*}. In *proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* 7278-7288 (2026)

Global underwater geolocation from time-lapsed polarization imagery. Sara Aghajanzadeh, **Xiaoyang Bai**, Zhongmin Zhu, David Forsyth, Viktor Gruev. In *proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* 6464-6473 (2026)

ERF-GS: Reconstructing fast motion from disjoint event-RGB viewpoints. **Xiaoyang Bai**[†], Zhenyang Li[†], Weiwei Xu, Edmund Y. Lam, Yifan Peng. In production for *Computer Visual Media Journal (CVMJ)* (2026)

E-WAVE: Event-based continuous optical flow via warping-aligned visual encoding. Jiale Wu[†], **Xiaoyang Bai**[†], Haoming Yu, Yiwei Chen, Yifan Peng, Weiwei Xu. *arXiv preprint arXiv:2609.34346* (2026).

SfM-free 3D Gaussian splatting from extremely sparse view. Zongqi He, Hanmin Li, Kin-Chung Chan, Yushen Zuo, Hao Xie, Zhe Xiao, Jun Xiao, **Xiaoyang Bai**, Yifan Peng, Kin-Man Lam. In *Computer & Graphics* 139 (2026)

EventTracer: Fast path tracing-based event stream rendering. Zhenyang Li[†], **Xiaoyang Bai**[†], Jinfan Lu, Pengfei Shen, Edmund Y. Lam, Yifan Peng. In *IEEE Transaction on Visualization and Computer Graphics (TVCG)* 32.8 (2026)

Structure-grounded training strategies aid generalization in stereo matching. Liangxun Ou, Yuhui Liu, Zhenyang Li, **Xiaoyang Bai**, Yifan Peng. In *proceedings of the 13th International Conference on 3D Vision (3DV)* 125-134 (2026)

Enhanced velocity field modeling for Gaussian video reconstruction. Zhenyang Li[†], **Xiaoyang Bai**[†], Tongchen Zhang, Pengfei Shen, Edmund Y. Lam, Yifan Peng. In *proceedings of IEEE International Symposium on Mixed and Augmented Reality (ISMAR)* 304-314 (2025)

Learning RGBD imaging via asymmetrically focused stereo cameras. Liangxun Ou, Yuhui Liu, **Xiaoyang Bai**, Yifan Peng. In *The Visual Computer* 41.15 (2025)

*: Corresponding author. †: Equal contribution.

Learning a global underwater geolocalization model with Sectoral Transformer. Xiaoyang Bai, Zhongmin Zhu, Alexander Schwing, David Forsyth, Viktor Gruev. In *Optics Express* 32.12 (2024)

Polarization-based underwater geolocalization with deep learning. Xiaoyang Bai, Zuodong Liang, Zhongmin Zhu, Alexander Schwing, David Forsyth, Viktor Gruev. In *eLight* 3.1 (2023)

Angle of polarization calibration for omnidirectional polarization cameras. Xiaoyang Bai, Zhongmin Zhu, Alexander Schwing, David Forsyth, Viktor Gruev. In *Optics Express* 31.4 (2023)

INVITED TALKS & CONFERENCE PRESENTATIONS

USTC-HKU Joint Research Seminar on AI-empowered Scientific Imaging & Visual Computing 08/2026

- (invited) “*Differentiable Rendering and 3D Reconstruction with Unconventional Visual Modalities*”

GAMES Technical Webinar, Seminar on Differentiable Rendering 06/2026

- (invited) “*Differentiable Rendering and 3D Reconstruction with Unconventional Visual Modalities*”

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 06/2026

- (technical paper presentation) “*PhyGaP: Physically-grounded Gaussians with Polarization Cues*”

IEEE International Symposium on Mixed and Augmented Reality (ISMAR) 10/2025

- (technical paper presentation) “*Enhanced Velocity Field Modeling for Gaussian Video Reconstruction*”

State Key Lab of CAD & CG, Zhejiang University 12/2024

- (invited) “*Unconventional Visual Information-Driven Deep Learning Practices*”

CSIG Youth Scientist Conference, Academic Service Forum 12/2024

- (invited) “*Unconventional Visual Information-Driven Deep Learning Practices*”

TEACHING & MENTORING

Teaching Assistant, University of Illinois at Urbana-Champaign 2018-2020

- CS 498 Applied Machine Learning
- CS 446 Machine Learning

Research Student Mentorship

- Mr. Zhenyang Li Ph.D. student of HKU
- Ms. Yuhui Liu Ph.D. student of HKU
- Mr. Liangxun Ou Ph.D. student of HKU
- Mr. Zongqi He Ph.D. student of HKU
- Mr. Jinfan Lu Ph.D. student of THU
- Mr. Haoming Yu Ph.D. student of ZJU
- Mr. Jiale Wu MEng student of ZJU
- Mr. Yeming Yao MPhil/UG student of HKU